

Devansh Parapalli

+91 87931 50182 | Hyderabad, TG, India | hey@parapalli.dev | github.com/DevParapalli | linkedin.com/in/devparapalli

PROFESSIONAL SUMMARY

AI/ML engineer with end-to-end production ownership across LLM inference infrastructure, agentic systems, and RAG pipelines. Collaborated with client stakeholders, audit teams and finance ops to engineer production systems for Cisco; from requirements through deployment and ongoing maintenance; serving 700+ monthly cases. Published AI/ML researcher (IJTE, 2025). Achieved MRR 0.75 / nDCG@5 0.7 over a 100K-chunk retrieval corpus. GCP Professional Cloud Architect certified.

WORK EXPERIENCE

Software Development Engineer

Jul 2025 — Present

Tata Consultancy Services Limited

Hyderabad, India

- **Client: Cisco;** SDefined requirements, owned architecture, and drove cross-functional alignment across Cisco support, FinOps, and audit teams as sole engineer across 3 concurrent production engagements.
 - Reduced case resolution time from **1+ day to under 30 seconds** for **700+ monthly tickets** by building **AIT CaseIQ** end-to-end: Classified repeating cases along with the support and FinOps teams, designed multi-intent detection, agentic tool-calling LLM workflows, and structured output delivery over a Kafka-backed event-driven ingestion architecture with auto-scaling containerized runners.
 - Automated SOX audit review for **500+ man-hours/month** of payroll reconciliation by engineering an LLM harness that ingests PAAT payroll records and Oracle General Ledger accruals, parses incoming email justifications, scores acceptability against policy, and tracks outstanding gaps; validated with finance and audit stakeholders, replacing manual review end-to-end.
 - Saved **800+ man-hours/month** across **4 FinOps workflows** by building async LangGraph-orchestrated pipelines in collaboration with FinOps and data teams: an SOP-driven multi-source report generator (Snowflake, MongoDB, OracleDB), a RAG + tool-calling knowledge chatbot, a natural language to multi-target query translator, and an automated data aggregation and Excel delivery pipeline.
- **Internal: GenAI Center of Excellence;** Led technical architecture and client demos across **7 RFPs**, contributing to \$2M+ in won engagements covering GenAI enablement, cloud migration, and legacy infrastructure modernization.
 - Eliminated external API dependency for enterprise model serving (**7B-120B parameters**) by architecting an on-premises **vLLM** and **LiteLLM** inference platform with centralized API gateway routing, PII detection, RBAC, and audit logging.

GenAI & Backend Development Intern

Jun 2024 — Aug 2024

PolymathAI

Nagpur, India

- Cut per-video AI processing time by **94% (3 min to 10 sec)** by designing a parallelized inference pipeline across horizontally-scaled containers, processing **14,600+ videos**, enabling real-time workflows.

PROJECTS

AIKO: AI-powered Knowledge Organizer (aiko.parapalli.dev)

Jun 2024 — Present

- Achieved **MRR 0.75** and **nDCG@5 0.7** over a **100K-chunk technical knowledge corpus** by implementing hybrid retrieval combining pgvector semantic search with keyword search fallback, with tuned chunk strategy and embedding model selection across multi-modal content; built to solve personal knowledge management at scale. Published in IJTE, March 2025.
- Sustained **p95 ingest latency of 3s** and **p99 of 10s** across a multi-modal document pipeline (PDFs, web content, multimedia) via a custom serialization layer with cloud-native and offline-capable deployment targets.
- Reduced LLM response latency by **25%** by replacing SSE with custom **WebSocket** duplex streaming for real-time token delivery.

ARA: AI-powered Research Assistant (ara.parapalli.dev)

Jan 2024 — May 2024

- Built a RAG-based research workspace with live web retrieval and per-prompt source citation across FastAPI, SvelteKit, Supabase, and containerized LLM runners; built as an exploration of RAG before widespread tooling existed.

EDUCATION

Government College of Engineering, Nagpur

Nagpur, India

B.Tech, Computer Science and Engineering

Dec 2021 — Jun 2025

CGPA: 8.28/10.0

TECHNICAL SKILLS

AI/ML: LLM Inference & Serving, vLLM, LiteLLM, Tool Calling, MCP, Agentic AI Systems, Transformers, Intent Detection, Semantic Ranking

Languages: Python, JavaScript/TypeScript, C/C++, SQL, Zig

Frameworks & Libraries: LangGraph, FastAPI, React.js/Next.js

Cloud & Infrastructure: GCP (Certified Architect), AWS (EC2, S3, EKS), Docker, Kubernetes, OpenShift, Terraform

Databases: PostgreSQL w/ pgvector, Spanner, MongoDB, OracleDB

Systems: Distributed Systems, Linux, Systems Programming

CERTIFICATIONS & ACHIEVEMENTS

- **GCP Professional Cloud Architect** (Dec 2025) | **GCP Cloud Digital Leader** | **GCP Generative AI Leader**
- **Publication:** "AI-based Knowledge Organizer for diverse data formats" - IJTE, March 2025